

Year 7 · Chapter 11 · Probability: answers (teacher copy)

11.01 Sample space (booklet p. 3)

Examples (model these live)

E1 List the sample space for rolling a die.
every face · {1, 2, 3, 4, 5, 6}

E2 List the sample space for tossing two coins.
coin 1, then coin 2 · {HH, HT, TH, TT}

E3 Are the outcomes on this spinner equally likely?
the sectors are different sizes · no

Questions (20 on today's skill, 5 from earlier lessons)

- | | | | |
|--------------------------|--|--|---|
| 1 {A, B, C, D, E} | 10 24 cm ² 8.06 | 16 H1, H2, H3, H4, H5, H6 · T1, T2, T3, T4, T5, T6 (12 outcomes) | 21 warmest: Jan (20°C) · 19 – 20 = -1°C 10.01 |
| 2 {S, C, H, O, L} | 11 2 | 17 No: there are different numbers of each colour. | 22 No: there are different numbers of each colour. |
| 3 {blue, white, green} | 12 {B, D, E} | 18 2, 3, 4, ..., 12 · No: 7 can happen 6 ways, 2 only 1 way | 23 ABC, ACB, BAC · BCA, CAB, CBA (6 orders) |
| 4 2, 4, 6, 8, 10, 12 | 13 No: there are different numbers of each colour. | 19 Swim 10.01 | 24 HH, HT, TH, TT: 4 · one of each: HT and TH, so 2 |
| 5 6 | 14 yes: the sectors are the same size | 20 No: there are different numbers of each colour. | 25 6 + 2 + 14 + 2 · = 24 10.01 |
| 6 {B, C, D} | 15 Swim 10, Soccer 2 · 10 – 2 = 8 10.01 | | |
| 7 {P, E, N, C, I, L} | | | |
| 8 {white, yellow, black} | | | |
| 9 2, 4, 6, 8 | | | |

11.02 Probability (booklet p. 5)

Examples (model these live)

E1 Find P(rolling an even number) on a die.
even: 2, 4, 6: 3 outcomes of 6 · $\frac{3}{6} = \frac{1}{2}$

E2 A bag has 3 red and 5 blue counters. Find P(red).
3 red out of 8 · $\frac{3}{8}$

E3 Find P(a heart) from a full deck of 52 cards.
13 hearts · $\frac{13}{52} = \frac{1}{4} = 25\%$

Questions (35 on today's skill, 11 from earlier lessons)

- | | | | |
|-------------------------|--|--|---|
| 1 $\frac{1}{6}$ | 14 3 out of 6 · $\frac{1}{2}$ | 25 1 out of 6 · $\frac{1}{6}$ | 36 $\frac{71}{12}$ 4.01 |
| 2 $\frac{1}{4}$ | 15 $\frac{3}{7}$ | 26 No: there are different numbers of each colour. 11.01 | 37 6 + 7 = 13 out of 14 · $\frac{13}{14}$ |
| 3 $\frac{1}{2}$ | 16 $\frac{2}{3}$ | 27 HH, HT, TH, TT · P(HH) = $\frac{1}{4}$, P(one of each) = $\frac{1}{2}$ | 38 5 ways out of 36 · $\frac{5}{36}$ |
| 4 $\frac{1}{6}$ | 17 2 out of 8 · $\frac{1}{4}$ | 28 $\frac{1}{52} \approx 1.9\%$ | 39 7 + 5 = 12 out of 18 · $\frac{2}{3}$ |
| 5 $\frac{1}{3}$ | 18 $\frac{1}{26} \approx 3.8\%$ | 29 1 ways out of 36 · $\frac{1}{36}$ | 40 No: there are different numbers of each colour. 11.01 |
| 6 $\frac{1}{3}$ | 19 H1, H2, H3, H4, H5, H6 · T1, T2, T3, T4, T5, T6 (12 outcomes) 11.01 | 30 2 + 2 = 4 out of 7 · $\frac{4}{7}$ | 41 1 out of 6 letters · $\frac{1}{6}$ |
| 7 $\frac{1}{6}$ | 20 2 out of 5 letters · $\frac{2}{5}$ | 31 yes: the sectors are the same size 11.01 | 42 A = $\frac{1}{2} \times 4.4 \times 14 \cdot = 30.8 \text{ m}^2$ 8.06 |
| 8 $\frac{8}{9}$ | 21 4 out of 8 · $\frac{1}{2}$ | 32 6 ways out of 36 · $\frac{1}{6}$ | 43 3 ways out of 36 · $\frac{1}{12}$ |
| 9 {S, C, H, O, L} 11.01 | 22 $\frac{1}{4} \approx 25\%$ | 33 (0, -5) 9.06 | 44 3 + 5 = 8 out of 16 · $\frac{1}{2}$ |
| 10 $\frac{5}{13}$ | 23 1 out of 6 letters · $\frac{1}{6}$ | 34 5 + 5 = 10 out of 17 · $\frac{10}{17}$ | 45 $\frac{3}{4}$ 4.01 |
| 11 $\frac{1}{3}$ | 24 $1\frac{1}{2}$ 4.01 | 35 5 ways out of 36 · $\frac{5}{36}$ | 46 1 out of 5 letters · $\frac{1}{5}$ |
| 12 $\frac{1}{6}$ | | | |
| 13 B 9.02 | | | |

11.03 The probability scale (booklet p. 7)

Examples (model these live)

E1 Describe a probability of 0.8 in words.
between $\frac{1}{2}$ and 1 · likely

E2 Describe the chance of rolling a 7 on a die.
a die only goes to 6 · impossible: P = 0

E3 Order from least to most likely: rolling an even number, rolling a 6, rolling less than 7.
 $\frac{1}{6}$, $\frac{1}{2}$, 1 · 6, even, less than 7

Questions (24 on today's skill, 7 from earlier lessons)

- | | | | |
|-----------------------|-----------------------|-----------------------------------|---|
| 1 likely | 7 unlikely | 12 unlikely | 17 {B, A, N} 11.01 |
| 2 impossible | 8 even chance | 13 4 10.07 | 18 $\frac{1}{6} = 0.1667$ · unlikely |
| 3 0 | 9 $\frac{5}{7}$ 11.02 | 14 $\frac{7}{8} = 0.875$ · likely | 19 rolling a 9 on a die; rolling a 6 on a die; tossing a head; rolling a number less than 5 |
| 4 likely | 10 unlikely | 15 even chance | |
| 5 certain | 11 1 | 16 certain | |
| 6 $\frac{1}{2}$ (0.5) | | | |

20 1 out of 6 letters $\cdot = \frac{1}{6}$
11.02

21 $\frac{5}{6} = 0.8333 \cdot$ likely

22 rolling a 9 on a die;
rolling a 6 on a die; tossing a
head; rolling a number less
than 5

23 $\frac{1}{10} = 0.1 \cdot$ unlikely

24 e.g. rolling a 1 on a die

25 {white, red, black} 11.01

26 Probabilities are always
from 0 to 1.

27 Probabilities are always
from 0 to 1.

28 2 out of 4 $\cdot = \frac{1}{2}$ 11.02

29 Probabilities are always
from 0 to 1.

30 $60 = 12 \times h \cdot h = 5 \text{ m}$ 8.12

31 Probabilities are always
from 0 to 1.

11.04 Experimental probability (booklet p. 9)

Examples (model these live)

E1 A coin is tossed 50 times and lands
heads 28 times. Find the experimental
P(heads).
 $\frac{28}{50} \cdot = 0.56$

E2 What is the theoretical P(heads)?
Compare.
 $\frac{1}{2} = 0.5 \cdot$ close to 0.56

E3 A die is rolled 120 times. How many
3s do you expect?
 $\frac{1}{6} \times 120 \cdot = 20$

Questions (32 on today's skill, 9 from earlier lessons)

1 $\frac{2}{5} = 0.4$

2 10

3 56%

4 $\frac{11}{20} = 0.55$

5 50

6 40%

7 $\frac{21}{50} = 0.42$

8 20

9 100

10 90%

11 $\frac{3}{5} = 0.6$

12 (0, 6) 9.06

13 experimental: $\frac{11}{30} \approx 0.37 \cdot$

theoretical: $\frac{1}{6} \approx 0.17$

14 40%

15 $\frac{13}{20} = 0.65$

16 $\frac{9}{11}$ 4.12

17 experimental: $\frac{1}{5} \approx 0.2 \cdot$

theoretical: $\frac{1}{6} \approx 0.17$

18 $\frac{1}{4} \times 40 \cdot = 10$

19 rolling a 9 on a die;
rolling a 6 on a die; tossing a
head; rolling a number less
than 5 11.03

20 experimental: $\frac{13}{60} \approx 0.22 \cdot$

theoretical: $\frac{1}{6} \approx 0.17$

21 $\frac{1}{4} \times 80 \cdot = 20$

22 $\frac{2}{3} \times 90 \cdot = 60$

23 experimental: $\frac{2}{15} \approx 0.13 \cdot$

theoretical: $\frac{1}{6} \approx 0.17$

24 {S, C, H, O, L} 11.01

25 26 out of 74 $\approx 0.35 \cdot$
probably not: 6 came up far
more often than the others

26 1 out of 7 letters $\cdot = \frac{1}{7}$
11.02

27 $\frac{1}{4} \times 120 \cdot = 30$

28 experimental: $\frac{1}{10} \approx 0.1 \cdot$

theoretical: $\frac{1}{6} \approx 0.17$

29 22 out of 74 $\approx 0.3 \cdot$
probably not: 6 came up far
more often than the others

30 The 100 tosses: more
trials usually give a result
closer to the true probability.

31 $\frac{2}{5} = 0.4 \cdot$ unlikely 11.03

32 24 out of 74 $\approx 0.32 \cdot$
probably not: 6 came up far
more often than the others

33 192 cm^2 8.07

34 The 100 tosses: more
trials usually give a result
closer to the true probability.

35 The 100 tosses: more
trials usually give a result
closer to the true probability.

36 27 out of 79 $\approx 0.34 \cdot$
probably not: 6 came up far
more often than the others

37 $1 \text{ m}^3 = 100 \times 100 \times 100 = 1$
 $000 \text{ 000 cm}^3 = 2 \text{ 000 000}$
 cm^3 8.10

38 $\frac{3}{5} \times 150 \cdot = 90$

39 $\frac{1}{8}$ 4.12

40 The 100 tosses: more
trials usually give a result
closer to the true probability.

41 28 out of 80 $\approx 0.35 \cdot$
probably not: 6 came up far
more often than the others

11.05 Complementary events (booklet p. 11)

Examples (model these live)

E1 Find P(not rolling a 6) on a die.
 $1 - \frac{1}{6} \cdot = \frac{5}{6}$

E2 P(rain) = 0.3. Find P(no rain).
 $1 - 0.3 \cdot = 0.7$

E3 What is the complement of rolling
an even number?
the event does not happen $\cdot$ rolling an
odd number

Questions (35 on today's skill, 14 from earlier lessons)

1 0.2

2 $\frac{5}{6}$

3 rolling 1, 2, 3, 4 or 5

4 0.1

5 $\frac{5}{6}$

6 picking a black card

7 0.5

8 $\frac{5}{6}$

9 $\frac{2}{3}$ 11.02

10 $\frac{5}{6}$

11 losing the game

12 0.6

13 80 cm^2 8.07

14 $100\% - 28\% \cdot = 72\%$

15 $\frac{5}{6}$

16 tossing a tail

17 certain 11.03

18 0.8

19 $100\% - 83\% \cdot = 17\%$

20 $P(\text{white}) = \frac{1}{3} \cdot P(\text{not}$
 $\text{white}) = 1 - \frac{1}{3} = \frac{2}{3}$

21 yes: the sectors are the
same size 11.01

22 $1 - \frac{5}{12} \cdot = \frac{7}{12}$

23 $100\% - 33\% \cdot = 67\%$

24 $\frac{3}{5} \times 150 \cdot = 90$ 11.04

25 $P(\text{white}) = \frac{4}{11} \cdot P(\text{not}$
 $\text{white}) = 1 - \frac{4}{11} = \frac{7}{11}$

26 $1 - \frac{2}{7} \cdot = \frac{5}{7}$

27 8 9.04

28 $100\% - 45\% \cdot = 55\%$

29 {white, green, red} 11.01

30 $0.2 + 0.3 = 0.5 \cdot P(\text{green}) =$
0.5

31 $P(\text{green}) = \frac{7}{20} \cdot P(\text{not}$
 $\text{green}) = 1 - \frac{7}{20} = \frac{13}{20}$

32 $\frac{2}{3}$ 11.02

33 $1 - \frac{4}{9} \cdot = \frac{5}{9}$

34 $0.3 + 0.4 = 0.7 \cdot P(\text{green}) =$
0.3

35 25 out of 50 $\cdot = 50\%$

36 $A = \frac{1}{2} \times 8.9 \times 10 \cdot = 44.5$
 m^2 8.06

37 No. Complements must
add to 1, and $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.

38 68% 11.04

39 $0.4 + 0.3 = 0.7 \cdot P(\text{green}) =$
0.3

40 29 out of 40 $\cdot = 72.5\%$

41 No: there are different
numbers of each colour. 11.01

42 37 out of 40 $\cdot = 92.5\%$

43 $0.3 + 0.5 = 0.8 \cdot P(\text{green}) =$
0.2

44 A 9.02

45 $1 - \frac{3}{8} \cdot = \frac{5}{8}$

46 6 11.01

47 15 out of 30 $\cdot = 50\%$

48 $0.3 + 0.3 = 0.6 \cdot P(\text{green}) =$
0.4

49 even chance 11.03